NO.7 CHARGING SYSTEM WARNING LIGHT ILLUMINATED/MESSAGE RELATED TO CHARGING SYSTEM MALFUNCTION INDICATED IN DISPLAY [SKYACTIV-D 1.5]

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7	CHARGING SYSTEM WARNING LIGHT ILLUMINATED/MESSAGE RELATED TO CHARGING SYSTEM MALFUNCTION INDICATED IN DISPLAY
DESCRIPTION	• Charging system warning light illuminated. • Message related to charging system malfunction indicated in display.
POSSIBLE CAUSE	• i-ELOOP system malfunction (with i-ELOOP) • Erratic signal to PCM — Generator or related circuit malfunction • Battery malfunction • Generator malfunction • Instrument cluster malfunction

Diagnostic Procedure

STEP	INSPECTION	RESULTS	ACTION
1	**VERIFY PCM DTC** • Retrieve PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	**With i-ELOOP:Without i-ELOOP:** • Perform the i-ELOOP symptom troubleshooting “No. 1 MESSAGE RELATED TO i-ELOOP SYSTEM FUNCTION INDICATED DISPLAY”, then go to Step 3. (See No.1 MESSAGE RELATED TO i-ELOOP SYSTEM MALFUNCTION INDICATED IN DISPLAY [i-ELOOP].) • Go to the next step.
2	**VERIFY CURRENT INPUT SIGNAL STATUS** **Caution** **• While performing this step, always operate the vehicle in a safe and lawful manner.** **• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — ALTT V • Do the PIDs indicate the correct values under the trouble condition? (See PCM INSPECTION [SKYACTIV-D 1.5].)	Yes	Inspect the related sensor and circuit. • If there is any malfunction: — Repair or replace the malfunctioning part according to the inspection results, then go to Step 6. • If there is no malfunction: — Go to the next step.
		No	Go to the next step.
3	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 6. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning part according to the inspection results, then go to Step 6. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
5	**INSPECT INSTRUMENT CLUSTER** • Inspect the instrument cluster. (See INSTRUMENT CLUSTER INSPECTION.) • Is the instrument cluster normal?	Yes	Go to the next step.
		No	Replace the instrument cluster, then go to the next step. (See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)

STEP	INSPECTION	RESULTS	ACTION
6	Verify the test results. • If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].) • If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis. — If the vehicle is repaired, troubleshooting is completed. — If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)		